

How disrupted spatial fluxes reshape food web structure along the river–sea continuum

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Abstract

Introduction

This is some text.

Connectivity

Key mechanisms

- Small dispersal stabilizes through spatial structure and asynchrony.
- Refuge effects occur when predator have lower efficiency when prey is rare (Holling type III).
- Prey switching of mobile top predators can stabilize connected food webs through ‘spatial coupling’.
- At high dispersal/connectivity the stabilizing effect wanes as strongly connected patches behave as a single patch.
- Studies between pairs of the following connectivity, complexity and stability are common. However, studies investigating how these are interconnected are rare.

Important references

- LeCraw 2014 - Review the effect of connectivity on food web complexity and stability.
- McCann 2005 - Consumers link food web in space, and their response can shape the spatial response of coupled food webs.
- Pillai 2011 - Metacommunity theory shows that food web complexity is maximized at intermediate level of colonization rates.

Dams

Key mechanisms

- Decrease river connectivity / reduce migration.
- Reservoir, lotic from lentic conditions, can result in hypoxic conditions. Favor phytoplanktons and macrophytes.
- Reservoir increase residence time of nutrients upstream and decrease the downstream output, which can result in a decrease in primary productivity.
- Hydropeaking - that is, the sudden release of large water to answer the higher electric demand - can harm riverine community due to sudden change in water level and water flow.
- Cumulative impact of dams can reduce the nutrient input in deltas.

Important references

- Ding 2026 - more endangered species near dam at a global scale.
- Dean 2023 - cumulative impact of downstream dam on potamodromous species.
- He 2024 - review dam impacts on riverine community.
- Pedersen 2012 - associate the drastic decline in the European eel to dam impacts.

Appendix

Model DAG

Put model DAG here using targets.